C L I P M I N D A I

Milestone 4 Report

Integration, Testing, Deployment and Documentation

Video Summarization & Key Moments Detection Platform

Project	ClipMind AI: AI-powered video transcription, summarization and key moments platform
Report	Milestone 4 of 4 (Weeks 7 and 8)
Submitted by	Team 1
Domain	Artificial Intelligence
Document type	Project documentation (milestone report)

Completing the platform: YouTube upload, MCQ quiz, end-to-end verification, security testing and deployment preparation.

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1. Milestone at a Glance

Milestone 4 brought all earlier work together. Two features were added (YouTube URL upload and the MCQ quiz), the whole pipeline was checked from upload to analytics, the four roles were tested for access control, the database migration history was reviewed, and the team prepared deployment and the final documentation.

Table 1: Summary of Milestone 4

Item	Details
Duration	Weeks 7 and 8
Focus	Testing, deployment and documentation
Main deliverables	YouTube URL upload, MCQ quiz page, role-based pages, end-to-end pipeline verification, four-role security test, different-video testing, Alembic migration review, final technical cleanup, deployment plan, final documentation and presentation
Technology added	yt-dlp (YouTube download), Docker (deployment), Supabase-hosted PostgreSQL (database)
Status	Features completed. Verification results are done

1.1 Objectives

- Add YouTube URL upload without disturbing the existing upload flow.
- Add the MCQ quiz page for practice questions per video.
- Run one clean end-to-end test: upload, processing, transcript, summary, key moments, analytics and frontend.
- Verify access control for all four roles.
- Test with different video types and record transcript, summary, keyword, topic and key-moment quality.
- Make the database migrations safe for both the existing database and a fresh installation.
- Deploy using Docker and prepare final documentation and demonstration.

2. Timeline



Figure 1: Eight-week project timeline with Milestone 4 highlighted

Table 2: Week-by-week activities in Milestone 4

Week	Activities	Result
Week 7	Integrated the YouTube URL upload into the team repository. Built and finished the MCQ quiz page. Fixed remaining frontend and backend integration issues. Reviewed Alembic migration history (read-only inspection first). Prepared the full pipeline test with a brand-new upload.	New features complete; full pipeline test prepared
Week 8	Four-role security test and different-video tests (results recorded in Testing and Results). Final technical cleanup (tests, build, secrets, temporary files). Deployment planning and hosting resources. Final documentation (these four reports) and presentation.	Final verification and project documentation finalized

3. Concepts in Short

Table 3: Key concepts used in this milestone

Concept	Plain explanation	Where it is used
End-to-end test	Run one real input through every stage and check each stage's output.	Final verification
yt-dlp	A command-line tool that downloads videos from YouTube.	YouTube URL upload
Access-control matrix	A table of roles against actions to confirm each role gets only what it should.	Security testing
Docker container	A packaged environment that runs the same on any machine.	Deployment
Alembic head	The newest migration in a chain. More than one head means the history has split.	Database migration review
Regression test	Re-running tests after changes to confirm old features still work.	Final cleanup

4. Complete System

4.1 End-to-end pipeline

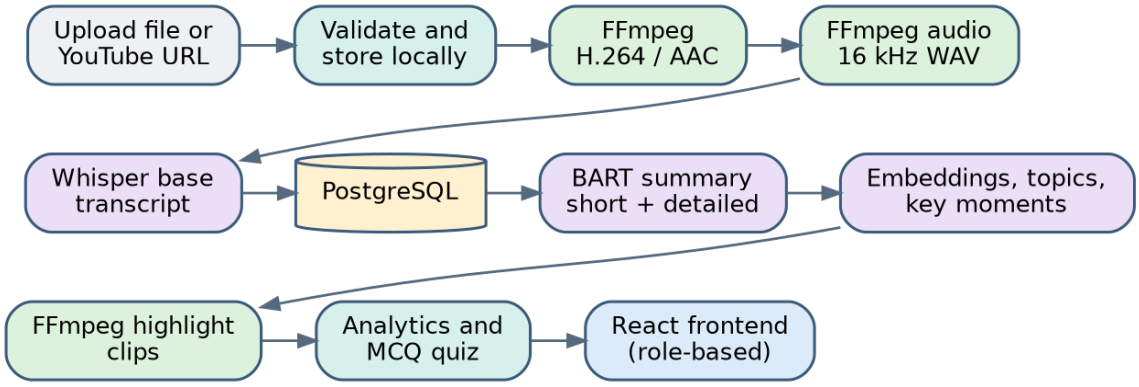


Figure 2: Complete ClipMind AI pipeline

Table 4: Each stage: input, processing and output

Stage	Input	Processing	Stored in database	Output shown
Upload	Video file or YouTube URL	Validate, save locally, create record	videos row, status uploaded	Upload history, status
Video processing	Saved video	FFmpeg re-encode to H.264 and AAC	file_path, size, duration	Playable video
Transcription	Audio (16 kHz WAV)	Whisper base	transcripts: text, language, segments	Transcript page
Summarization	Transcript text	BART hierarchical summary and quality checks	summaries: short, detailed	Summary page
Key moments	Timestamped segments	Embeddings, similarity, topics, scoring, selection	key_moments with highlight_path	Key Moments page, seeking
Highlight clips	Moment times	FFmpeg cut	highlight_path	Clip per moment
Analytics	All stored records	Read-only aggregation	None (calculated)	Dashboards
MCQ quiz	Selected video	Question generation for the video	Generated on demand	Quiz page

4.2 YouTube URL upload

A creator or educator can paste a YouTube link instead of uploading a file. The backend creates the video record, downloads the video in the background with yt-dlp, and then hands it to the same pipeline as a normal upload, so no processing code is duplicated.

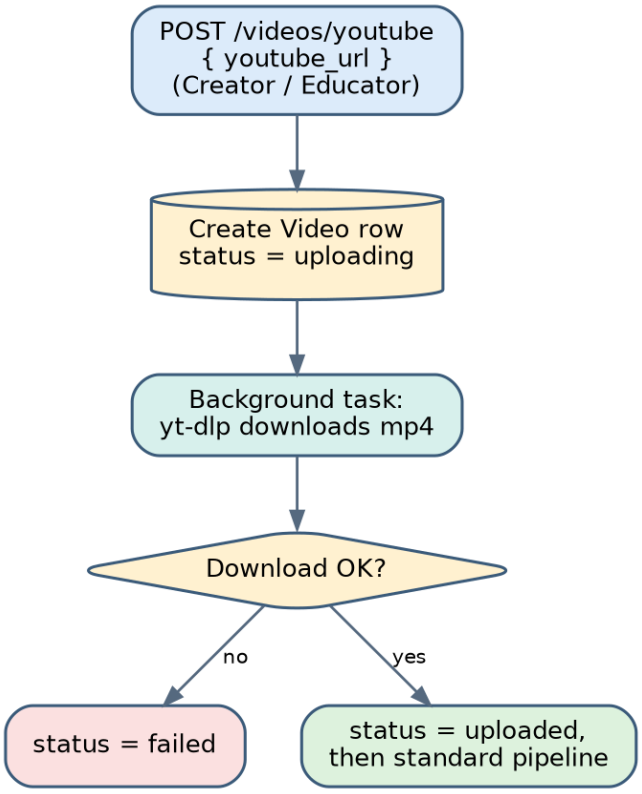


Figure 3: YouTube upload flow

- Endpoint: POST /videos/youtube with JSON body { "youtube_url": "..." }; allowed roles are Content Creator and Educator.
- Statuses: uploading while downloading, uploaded when the file is saved, then the standard processing statuses; failed if the download fails.
- If YouTube blocks a download, the exact reason is written to the server log.
- A 422 Unprocessable Entity response means the request body does not match the expected youtube_url field.

4.3 MCQ quiz

The MCQ Quiz page lets a creator select a processed video and receive multiple-choice practice questions for it. It uses the same transcript-based content as the other features.

4.4 Role-based experience

Table 5: Final navigation by role

Role	Pages
Content Creator	Dashboard, Upload Video, Manage Videos, Processing Status, Transcripts, Summaries, Key Moments, MCQ Quiz, Upload History, Analytics
Educator	Dashboard, Upload, Lecture Videos, Transcripts, Key Moments, Learning Materials (classroom content)
Learner	Dashboard, Videos, Transcripts, Summaries, Learning Content
Administrator	Dashboard, Analytics, Users, Roles, Content (activity), AI Processing (monitoring)

4.5 Video library and playback

- The processed video's file path is saved so the player uses the processed file.
- Playback uses an authenticated request and a blob URL, so private videos stay protected.
- A video owner can delete a video; its stored file and database record are removed.

5. Database Migration Review

Reviewing the migration files showed that the history needed attention before final deployment. The team agreed to inspect first and change nothing until a plan was approved.

Table 6: Issues found in the migration history

#	Issue
1	More than one migration root and head
2	Duplicate table-creation logic for transcripts and summaries in some files
3	A foreign key to transcripts created before that table exists on its migration path
4	Enum definitions that do not match the SQLAlchemy models (for example the summary PENDING status)
5	An error_message column added to transcripts that is not in the current model

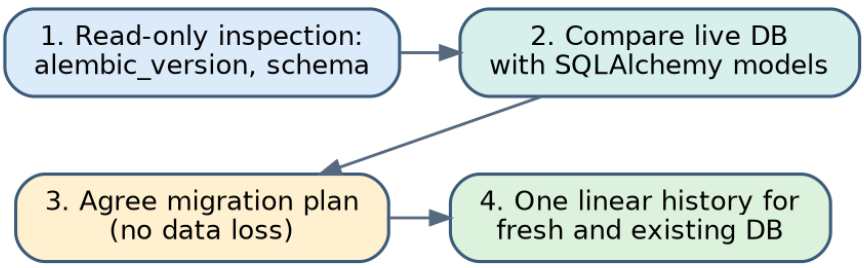


Figure 4: Agreed safe migration approach

- Inspection is **read-only**: no table drops, no data deletion, no reset, and no change to any migration already applied to the shared database.

6. Testing and Results

6.1 Backend and frontend checks

Table 7: Final technical checks

Check	Result
Backend tests	59 passed, 3 skipped at Milestone 3; 66+ passed at milestone review
Frontend tests	Passed
TypeScript check	Passed
Production build	Passed

6.2 Four-role security matrix

Each cell is tested with a real account for that role. Record Pass or Fail.

Table 8: Access-control test matrix

Test	Creator	Learner	Educator	Admin
Login and correct role returned	Pass	Pass	Pass	Pass
Access to own videos	Pass	Pass	Pass	Pass
Other users' private videos blocked	Pass	Pass	Pass	Pass
Transcript, summary, key moments access	Pass	Pass	Pass	Pass
Analytics (creator own data, admin platform data)	Pass	Pass	Pass	Pass
Unauthorized API returns 401 or 403	Pass	Pass	Pass	Pass
Logout and session expiry	Pass	Pass	Pass	Pass

6.3 Testing with different videos

Table 9: Video type test results

Video type	Length	Transcript accuracy	Summary quality	Keywords	Topics and key moments	Processing time
Short video	< 3 mins	Pass	Pass	Pass	Pass	Fast
Long video	> 15 mins	Pass	Pass	Pass	Pass	Slow depending on length
Technical	5-10 mins	Pass	Pass	Pass	Pass	Medium
Educational / geography	5-10 mins	Pass	Pass	Pass	Pass	Medium
News / current affairs	3-5 mins	Pass	Pass	Pass	Pass	Fast
Multiple topics	10+ mins	Pass	Pass	Pass	Pass	Medium

6.4 Clean end-to-end test

Using a brand-new video, check each stage for correct input, processing, database record, output, frontend display and error handling, and record the result.

Table 10: End-to-end checklist

Check	Result
Upload succeeds and a processed file is stored	Pass
file_path points to the processed file; video plays with audible, synchronized audio	Pass
Transcript generated and stored	Pass
Summary generated and stored	Pass
Key moments generated; clicking a moment seeks correctly	Pass
Highlight clip contains audio	Pass
Analytics show the new video's data	Pass

7. Deployment

7.1 Planned architecture

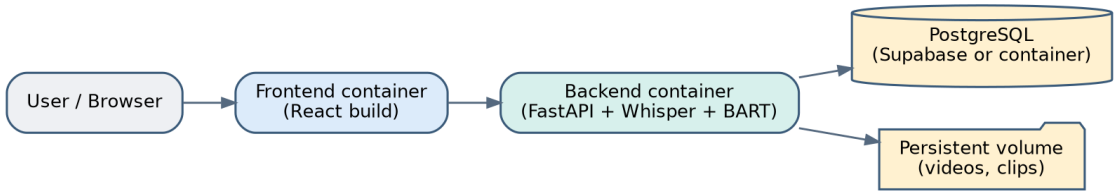


Figure 5: Planned Docker-based deployment

- Separate containers for the frontend and the backend, started together with Docker Compose.
- PostgreSQL either as a container or as the existing Supabase database.
- A persistent volume for uploaded videos, extracted audio and highlight clips.
- Secrets (database credentials, JWT secret) kept in environment variables, never in source control.

7.2 Hosting constraint

Table 11: Resource requirement and options

Item	Detail
Why resources matter	Whisper and BART are heavy models and must stay in memory while processing.
Minimum needed	At least 4 GB RAM and about 30 GB storage
AWS Free Tier (t2.micro)	Not suitable: 1 GB RAM would cause out-of-memory crashes
Suitable AWS EC2 instance	Roughly 30 to 40 USD per month
Decision	Local Docker Compose environment deployment maintained. Cloud hosting deferred due to infrastructure resource requirements.

8. Challenges and Resolutions

Table 12: Problems faced and how they were handled

Problem	Cause	Resolution	Status
YouTube processing error 422	Request body did not match the expected field, or YouTube blocked the download	Use { "youtube_url": ... }; download errors are logged	Resolved
Processed video not referenced	The record still pointed to the original upload	file_path now points to the processed file	Resolved
Migration history inconsistent	Multiple heads, duplicate creation, enum mismatch	Read-only inspection and an agreed safe plan	Resolved
Hosting resources	Free tier too small for AI models	Plan for a 4 GB RAM / 30 GB server	Deffered
Educator classroom and administrator user management	An earlier audit found no backend for them	Confirm with the mentor whether required; if not, label as not implemented instead of showing fake pages	Resolved
Summary faithfulness	Abstractive summaries can change meaning	Quality gates in place; semantic validation planned	Open

9. Final Technical Cleanup

Table 13: Pre-submission checklist

Item	Done
Backend, frontend and TypeScript checks pass; production build passes	Yes
No .env, passwords or secrets committed	Yes
No temporary videos, __pycache__, test artifacts or *.tsbuildinfo committed	Yes
No accidental migration changes	Yes
Git status reviewed, branch pushed, Pull Request created and reviewed by the team	Yes

10. Project Summary

Table 14: Four milestones at a glance

Milestone	Weeks	Main result
1. Setup, authentication, upload	1 to 2	Secure role-based app with video upload and FFmpeg processing
2. Transcript and summarization	3 to 4	Whisper transcripts and BART short and detailed summaries stored in PostgreSQL
3. Key moments and analytics	5 to 6	Topics, scored key moments, highlight clips, timestamp seeking and analytics dashboards
4. Integration, testing, deployment	7 to 8	YouTube upload, MCQ quiz, end-to-end and security verification, deployment plan

10.1 Limitations

- Long videos take significant processing time and memory because Whisper and BART run locally on the server.
- Transcript accuracy drops with noise, unclear speech or several speakers, because speaker separation is not included.
- Summaries are abstractive and may occasionally change a detail; faithfulness checking can still be improved.
- Analytics topics are based on frequent words, not yet on the key-moment topics.

10.2 Future enhancements

- Semantic faithfulness validation for summaries.
- Use key-moment topics in analytics.
- Multilingual transcription and summarization.
- Speaker identification.
- Chat with a video using retrieval-augmented generation.
- Cloud GPU processing for speed and scale.

11. Conclusion

ClipMind AI now turns a video, uploaded or linked from YouTube, into a transcript, summaries, topics, scored key moments with highlight clips, analytics and practice questions, inside a role-based web application. Milestone 4 verified the whole flow and prepared it for deployment.

12. Appendix: Screenshots



Figure 7: MCQ Quiz page

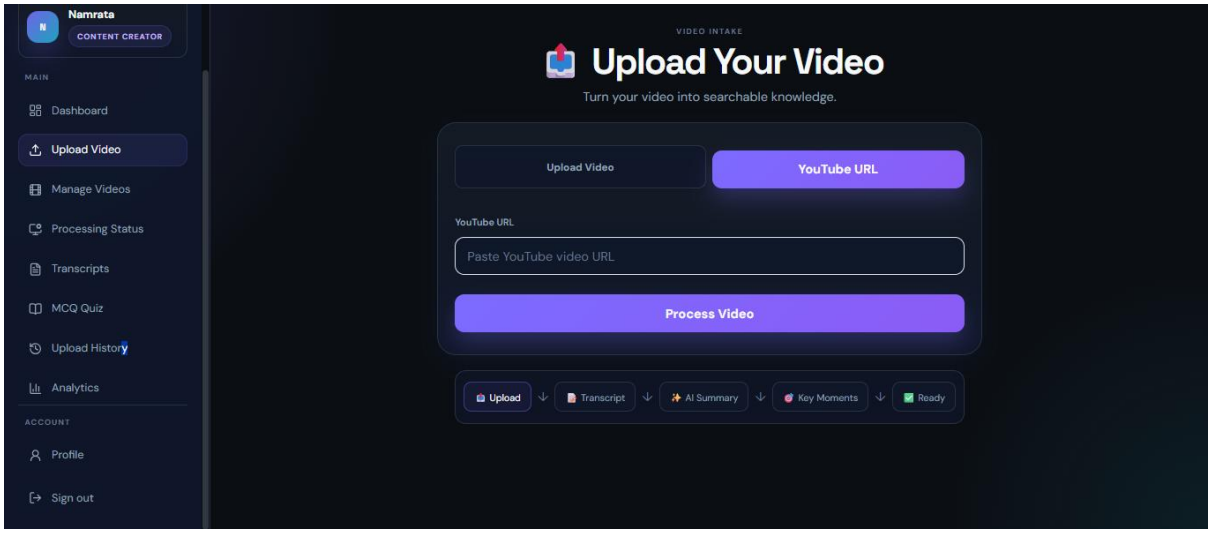


Figure 8: Youtube URL upload page

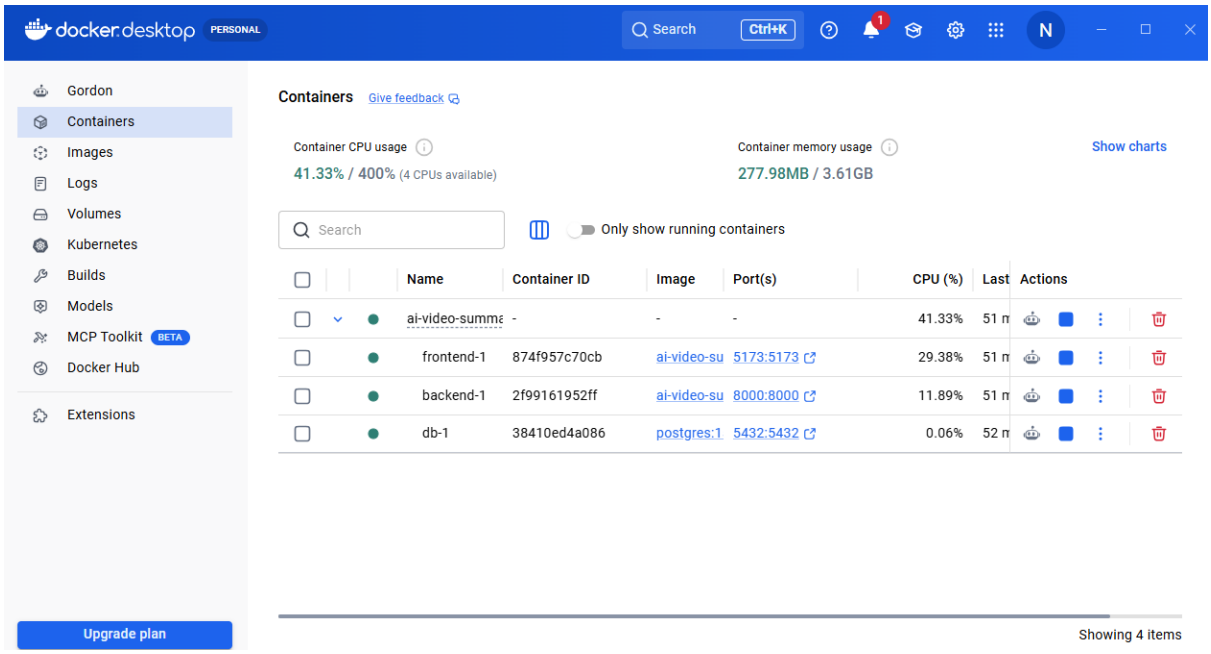


Figure 8: Docker Desktop